



Technische
Universität
Dresden



Deep Learning II

Digitization and Data Analytics

Structure of the Lecture Series

Lecture 1: Introduction

Lecture 2: Data Preprocessing

Lecture 3: Research Data Management

Lecture 4: Machine Learning I - Unsupervised Learning (as video)

Lecture 5: Machine Learning II - Supervised Learning (as video)

Lecture 6: Machine Learning III

Lecture 7: Data Analytics and Statistics

Lecture 8: Deep Learning I (as video)

Lecture 9: Deep Learning II

Lecture 10: Distributed Computing I

Lecture 11: Distributed Computing II

Lecture 12: Consequences

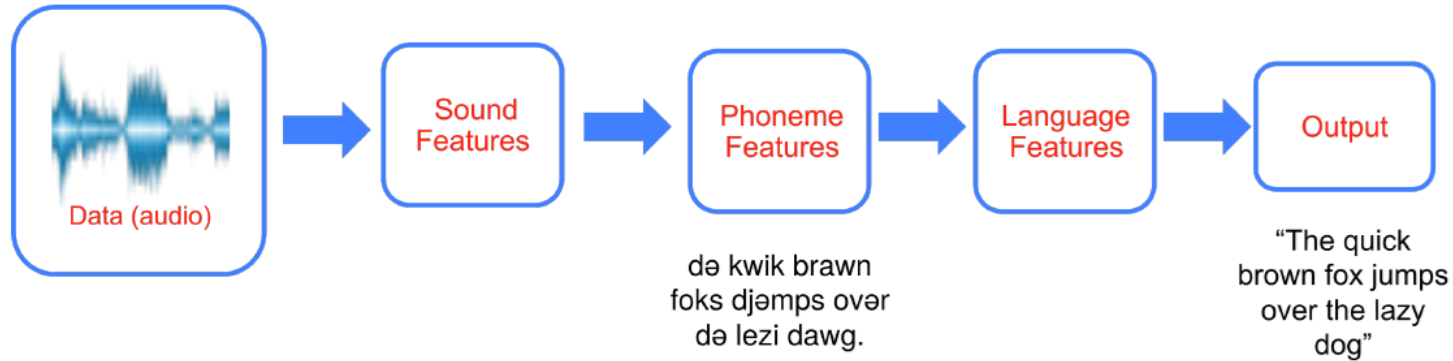
Structure of this Lecture

Network Architectures and Applications

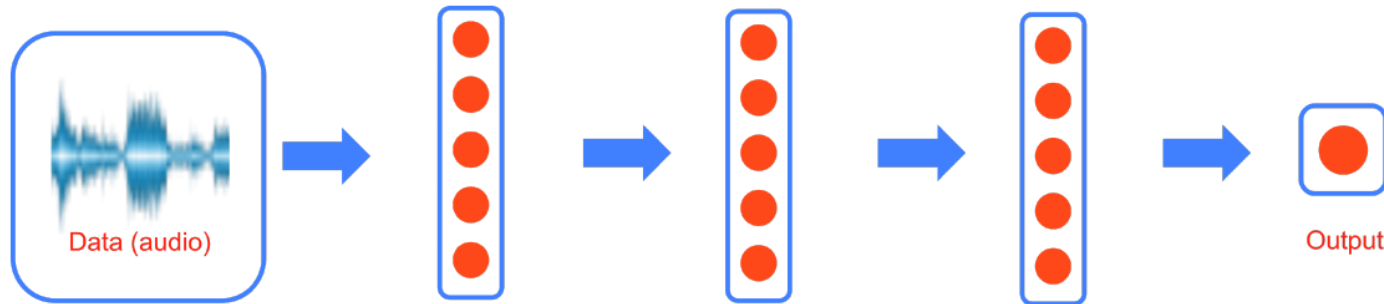
- The power of DNNs
- Images and Convolutional Neural Networks
- Sequences and Recurrent Neural Networks
- Text data and Transformers

Speech Recognition

Speech recognition used to employ hand-crafted layers of abstraction

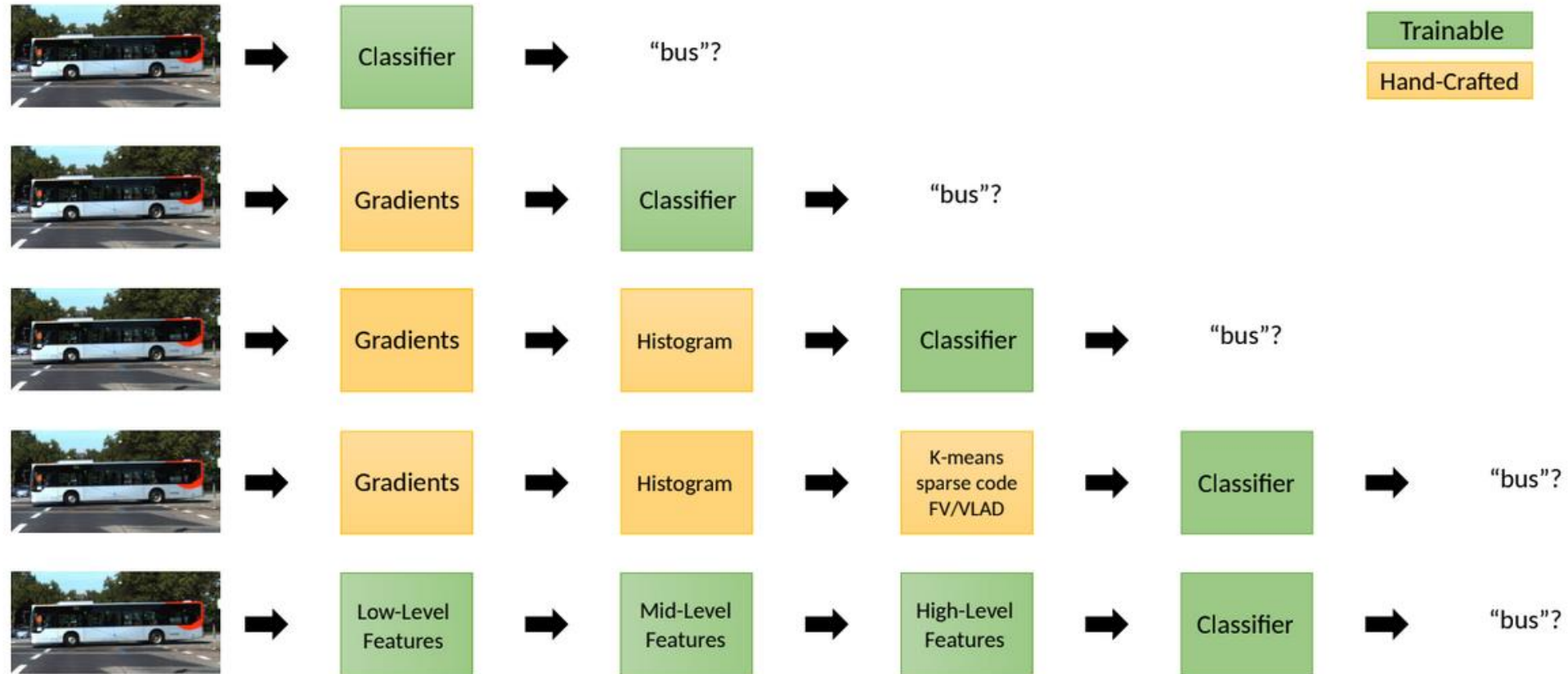


DNN learns these layers automatically, given enough data



Figures: Andrew NG @ GTC2015, Jeffrey Kelling HZDR

Image Classification



Public Datasets for Image Classification

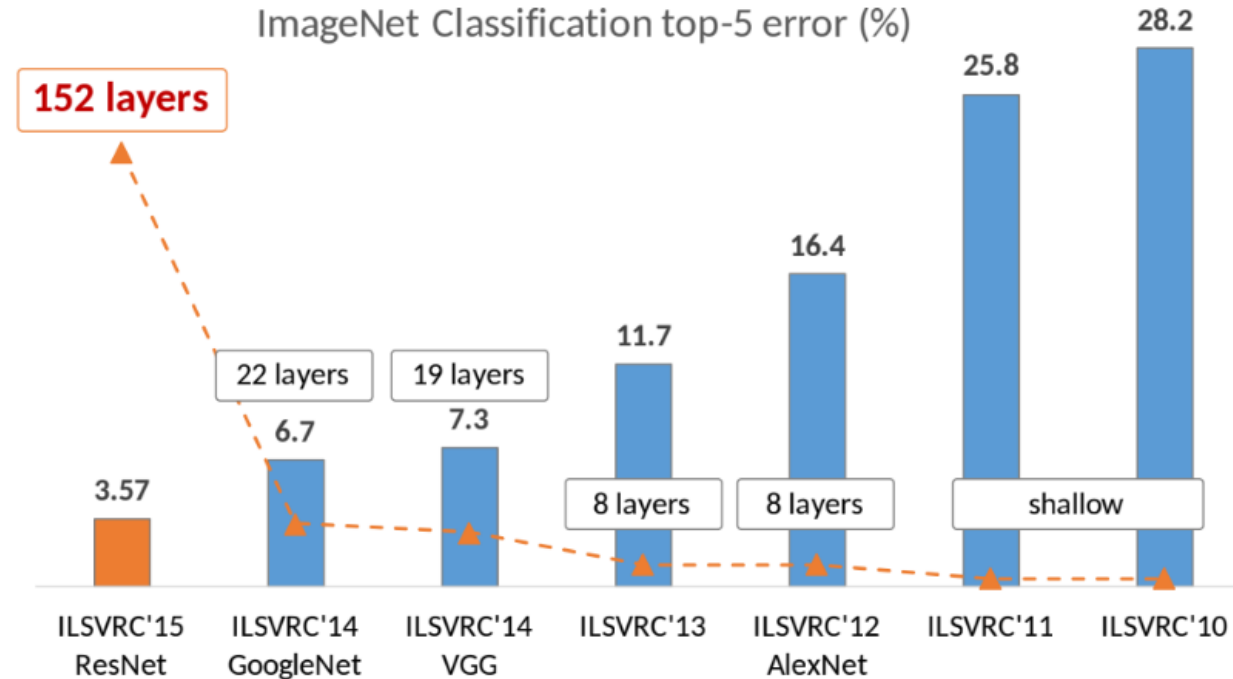
	MNIST	CIFAR-10	CIFAR-100	ImageNet
				
#classes	10	10	100	1000
year	1998	2009		2010

ImageNet challenge [1,2] also contains object localization and scene classification

[1] <http://image-net.org>

[2] http://rodrigob.github.io/are_we_there_yet/build/classification_datasets_results.html

ImageNet Large Scale Visual Recognition Challenge



Classification into 1000 object categories

Russakovsky et al.: ImageNet Large Scale Visual Recognition Challenge. IJCV, 2015.

<https://uni-tuebingen.de/fakultaeten/mathematisch-naturwissenschaftliche-fakultaet/fachbereiche/informatik/lehrstuehle/autonomous-vision/lectures/deep-learning/>

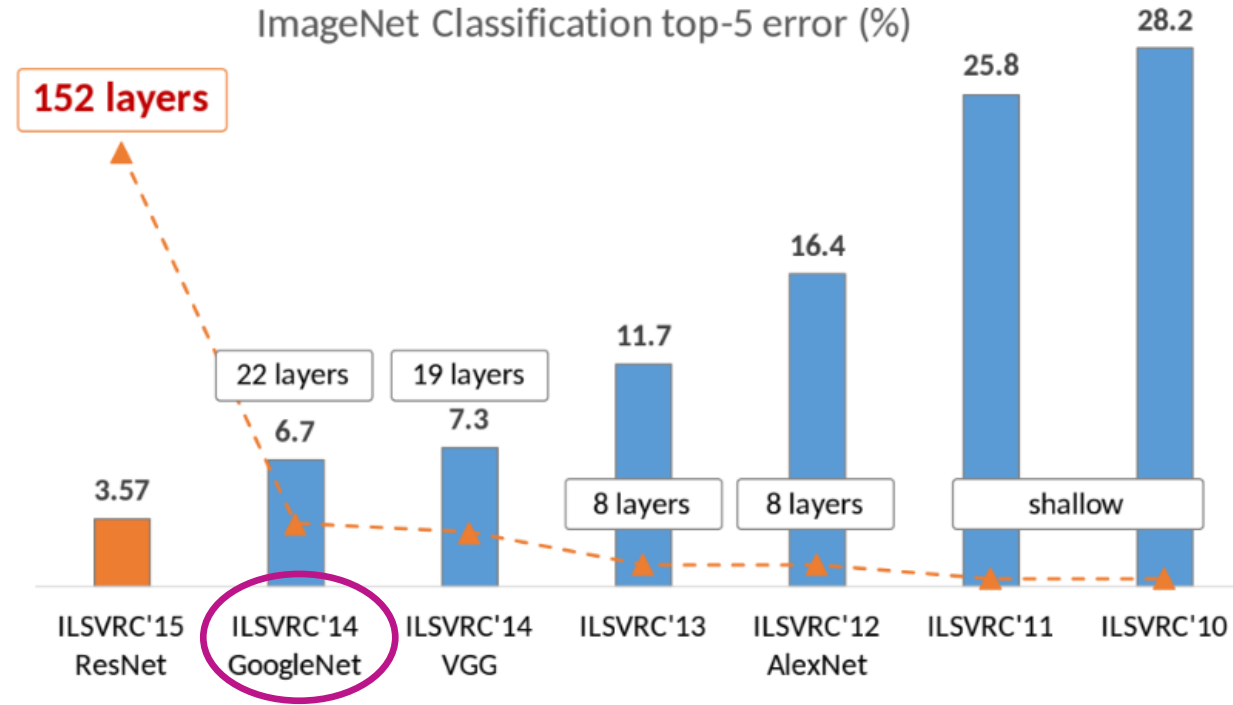
Lecture 07 Convolutional Neural Networks

Structure of this Lecture

Network Architectures and Applications

- The power of DNNs
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ImageNet Large Scale Visual Recognition Challenge



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Lecture 07 Convolutional Neural Networks

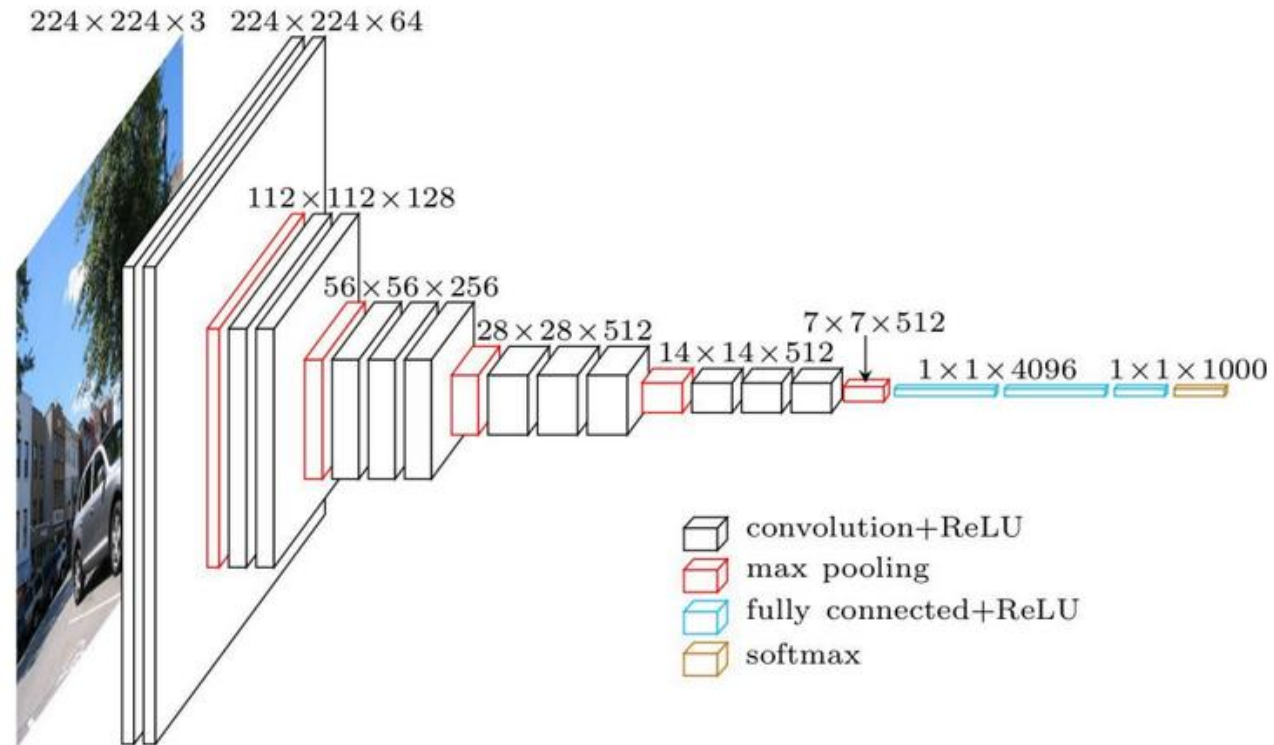
VGG* Network

Types of layers

Convolution

Pooling

Fully connected



* (Visual Geometry Group Oxford University)

[1] <https://www.kaggle.com/code/blurredmachine/vggnet-16-architecture-a-complete-guide/notebook>

Convolutional Networks - Notation

B : **Batch size**

W : **Width** of image or feature map

H : **Height** of image or feature map

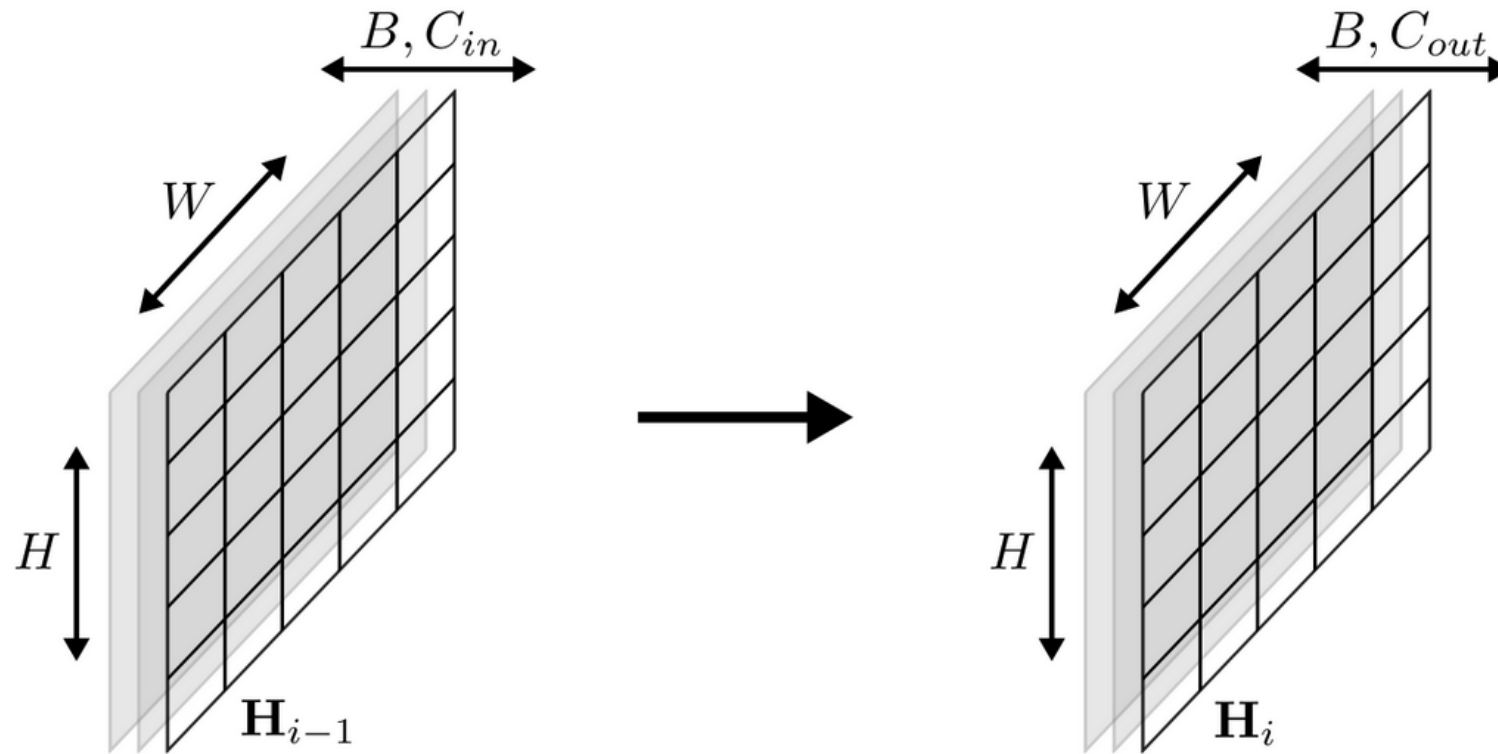
C : Number of **feature channels** (= neurons per pixel)

- C_{in} : Number of **feature** channels in **current** layer
- C_{out} : Number of **feature** channels in **next** layer
- $C_{in} = 1$ for first layer if input is a **grayscale** image

$C_{in} = 3$ for first layer if input is a color image

$K \times K$: **Convolutional filter kernel size**

Convolutional Layer – Spatial Dimension

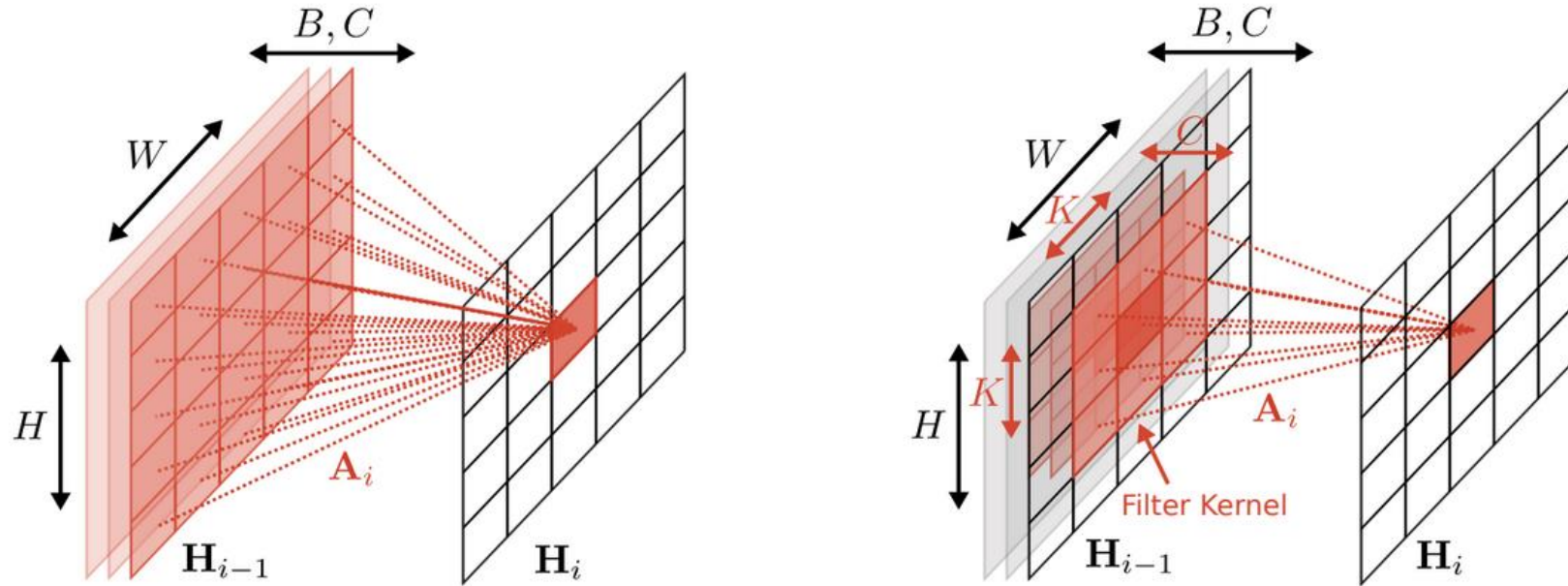


Note: In **contrast** to **MLPs**, Convolutional layers have a **spatial extent**.

<https://uni-tuebingen.de/fakultaeten/mathematisch-naturwissenschaftliche-fakultaet/fachbereiche/informatik/lehrstuehle/autonomous-vision/lectures/deep-learning/>

Lecture 07 Convolutional Neural Networks

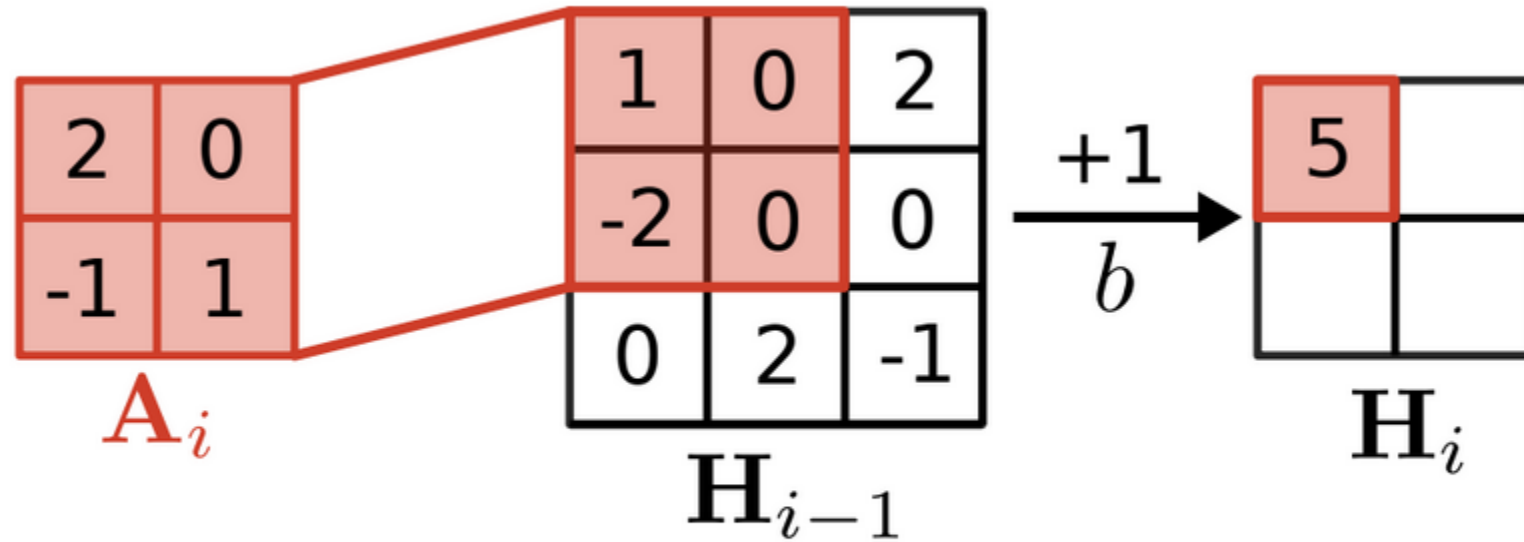
Convolutional Layer – Weights



- **Fully connected layer:** # Weights = $W \times H \times C_{out} \times (W \times H \times C_{in} + 1)$
- **Convolutional layer:** # Weights = $C_{out} \times (K \times K \times C_{in} + 1)$ („weight sharing“)
- C_{in}, C_{out} : input and output channels, $W \times H$: layer size, $K \times K$: kernel size
- Convolutional **kernel** is **three-dimensional**. For clarity, in the figure only one output channel is depicted.

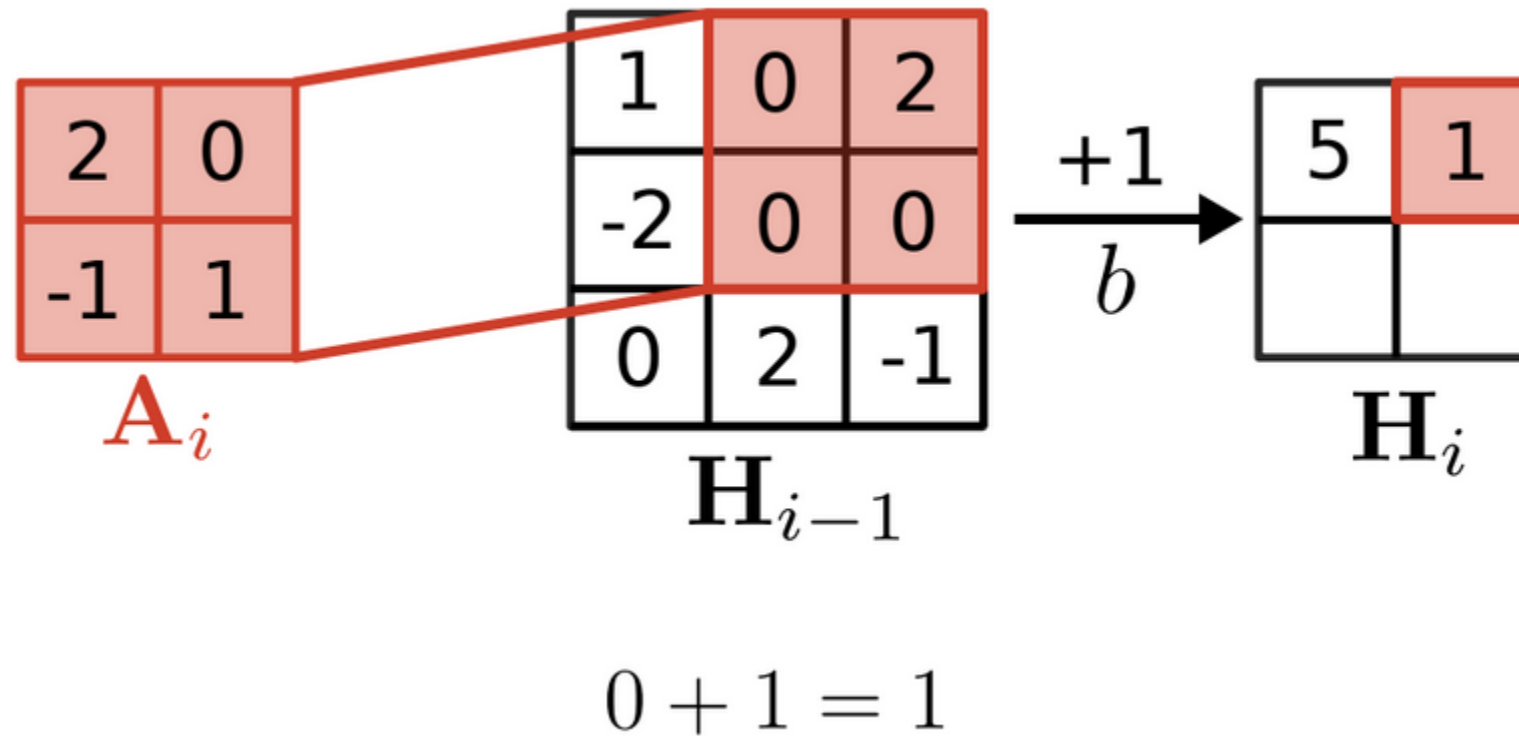
<https://uni-tuebingen.de/fakultaeten/mathematisch-naturwissenschaftliche-fakultaet/fachbereiche/informatik/lehrstuehle/autonomous-vision/lectures/deep-learning/>

Convolution Example

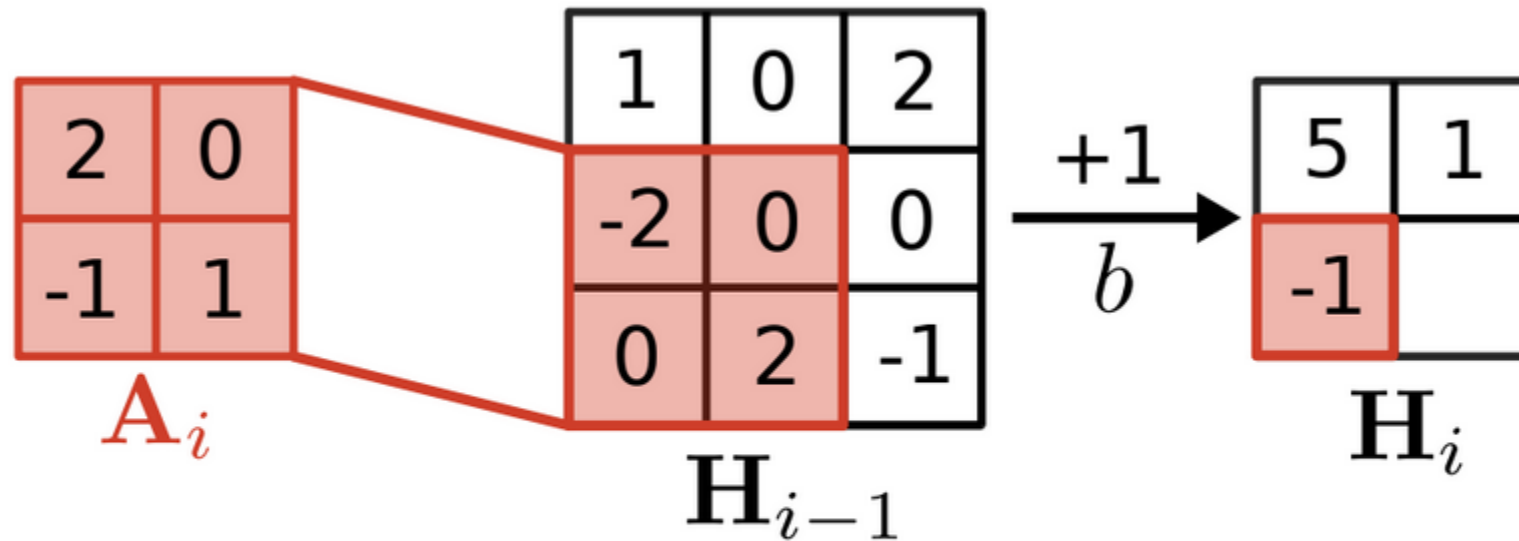


$$2 \cdot 1 + (-1) \cdot (-2) + 1 = 5$$

Convolution Example

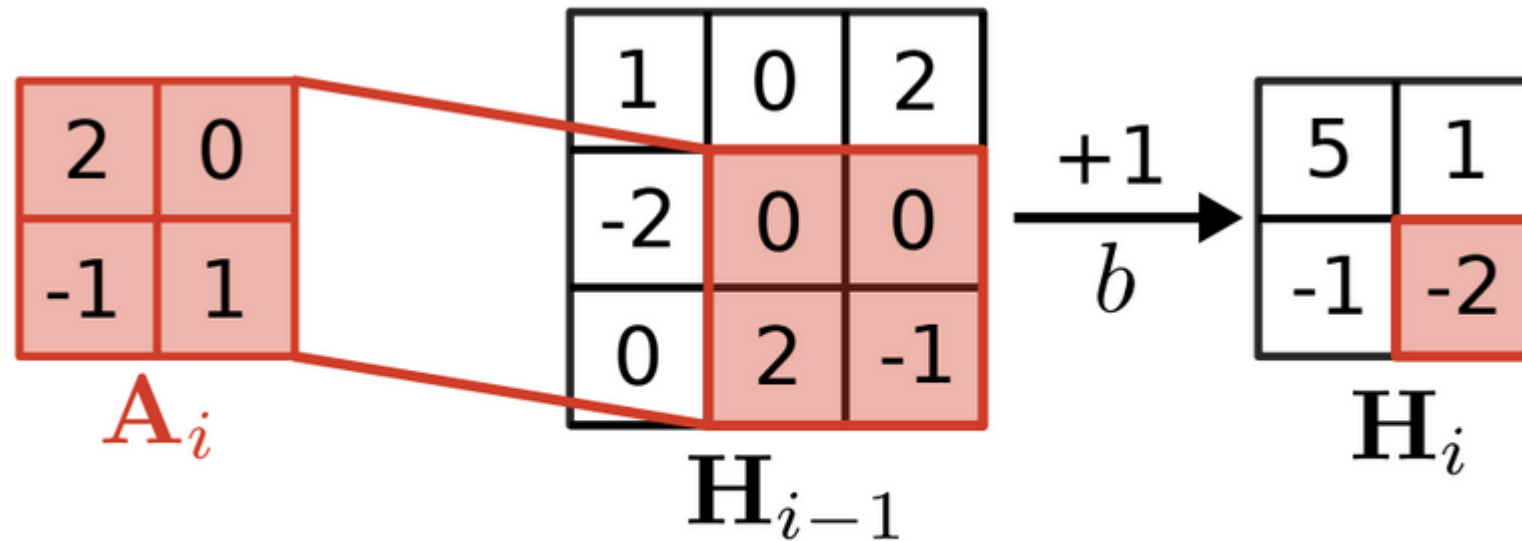


Convolution Example



$$2 \cdot (-2) + 1 \cdot 2 + 1 = -1$$

Convolution Example



$$(-1) \cdot 2 + 1 \cdot (-1) + 1 = -2$$

Structure of this Lecture

Network Architectures and Applications

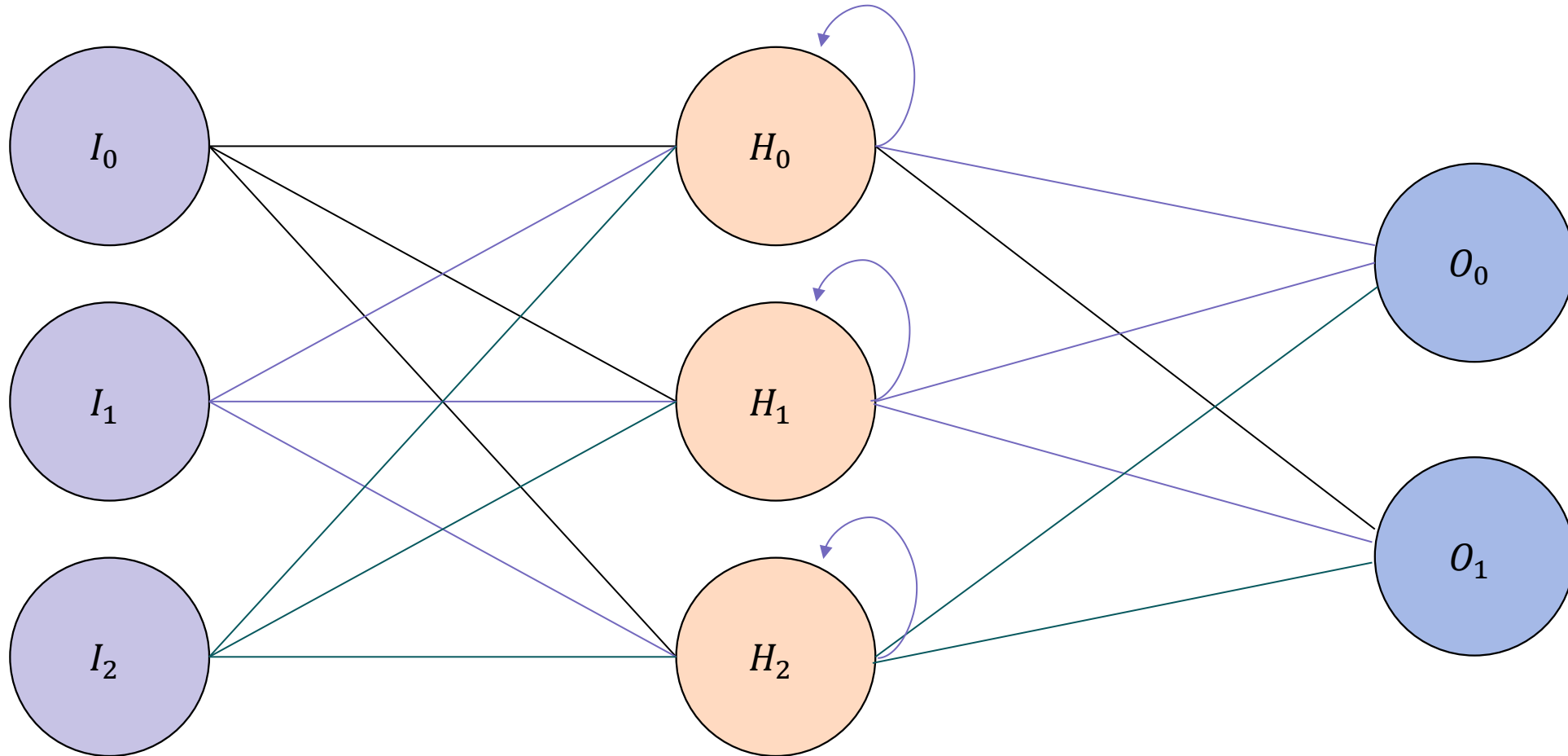
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Sequences of Variable Length - Applications

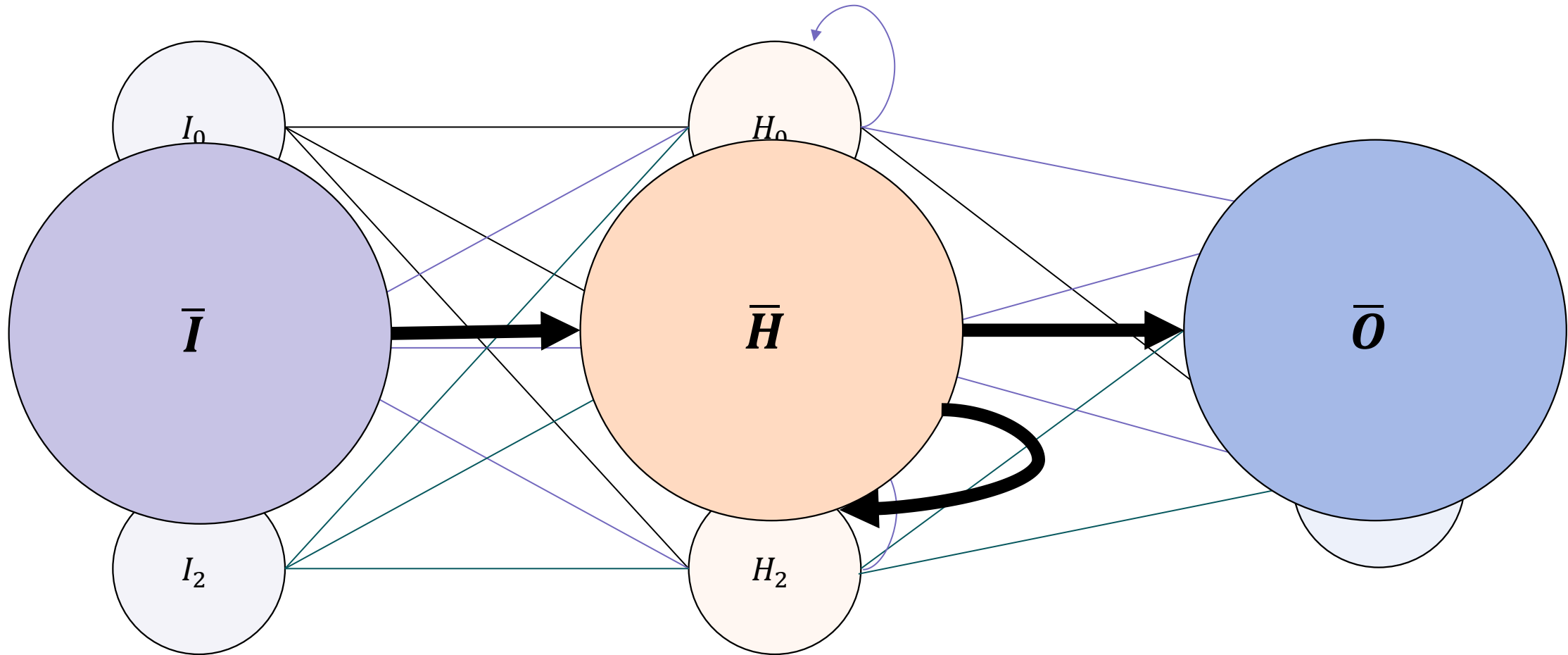
- Multiple object recognition
- Image segmentation
- Object tracking
- Image generation
- Image annotation
- Image captioning
- Machine translation
- Text generation

Each time step of process depends on „history“ (sequence of variable length)

Recurrent Neural Networks

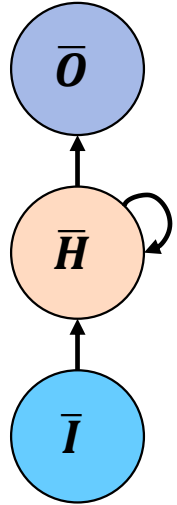


Recurrent Neural Networks

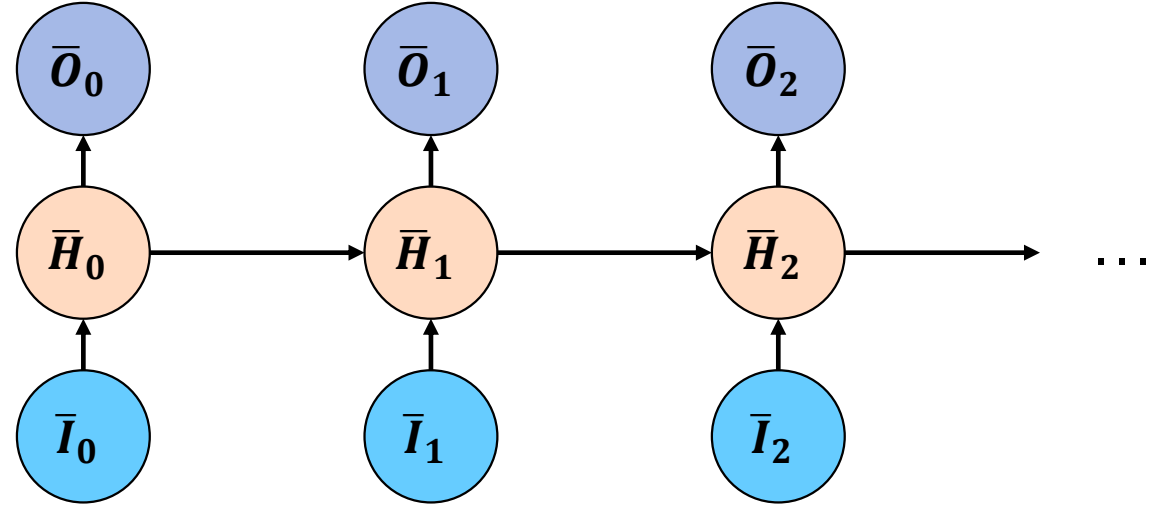


Recurrent Neural Networks (RNN)

RNN with **feedback** connection



RNN **unrolled** over time (index = time t)



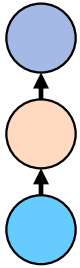
Update hidden state H based on **input and previous hidden state** using **same update rule** (**same/shared weights** and **activation** function) at each **time step**

Allows for processing sequences of **variable length** e.g. text data, not only fixed-sized vectors

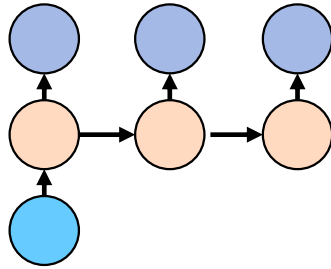
Infinite memory: H is function of all previous inputs (long-term dependencies)

Recurrent Neural Networks – Mapping Types

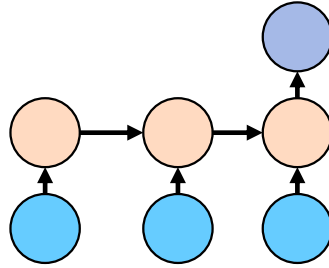
One to One



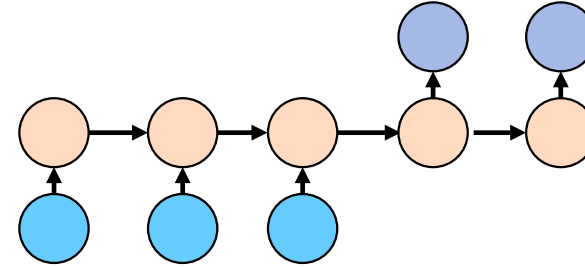
One to Many



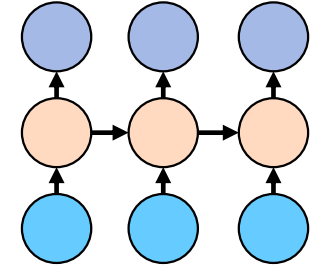
Many to One



Many to Many



Many to Many



RNNs allow for processing variable length inputs and outputs:

One to Many: Image captioning (image to sentence, music generation)

Many to One: Action recognition (video to action)

Many to Many: Machine translation (sentence to sentence)

Many to Many: Object tracking (video to object location per frame)

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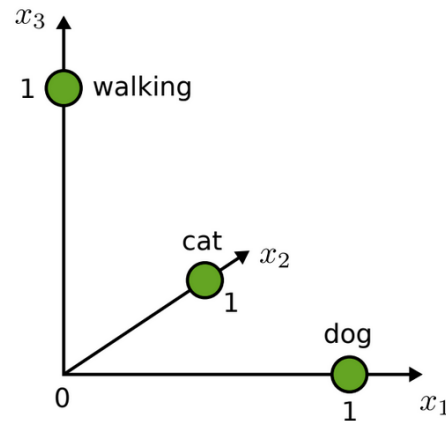
Neural Probabilistic Language Models – Word Representation

Dictionary $D = \{a, cat, dog, have, walking\}$

One-hot encoding

$a, cat, dog, have, walking$

1	0	0	0	0
0	1	0	0	0
0	0	1	0	0
0	0	0	1	0
0	0	0	0	1

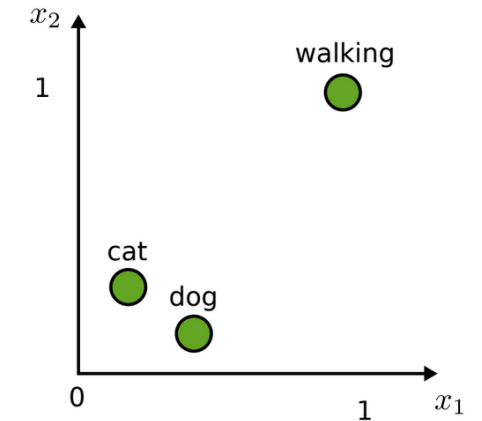


- sparse
- high-dimensional
- hard-coded

Word-embedding

$a \quad cat \quad dog \quad have \quad walking$

0.8	0.03	0.15	0.05	0.4
0.05	0.8	0.2	0.5	0.3
0.06	0.15	0.6	0.4	0.25
0.09	0.02	0.05	0.15	0.05

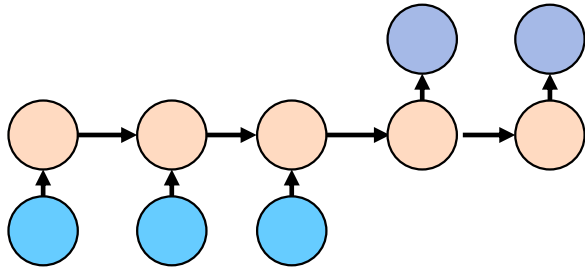


- dense
- lower-dimensional
- learned from data

Bengio et al.: A Neural Probabilistic Language Model, Journal of Machine Learning Research, 2003

Text Data – Processing Sequences of Words

RNN



Self-Attention [1]

relates **all tokens of a sequence** with each other

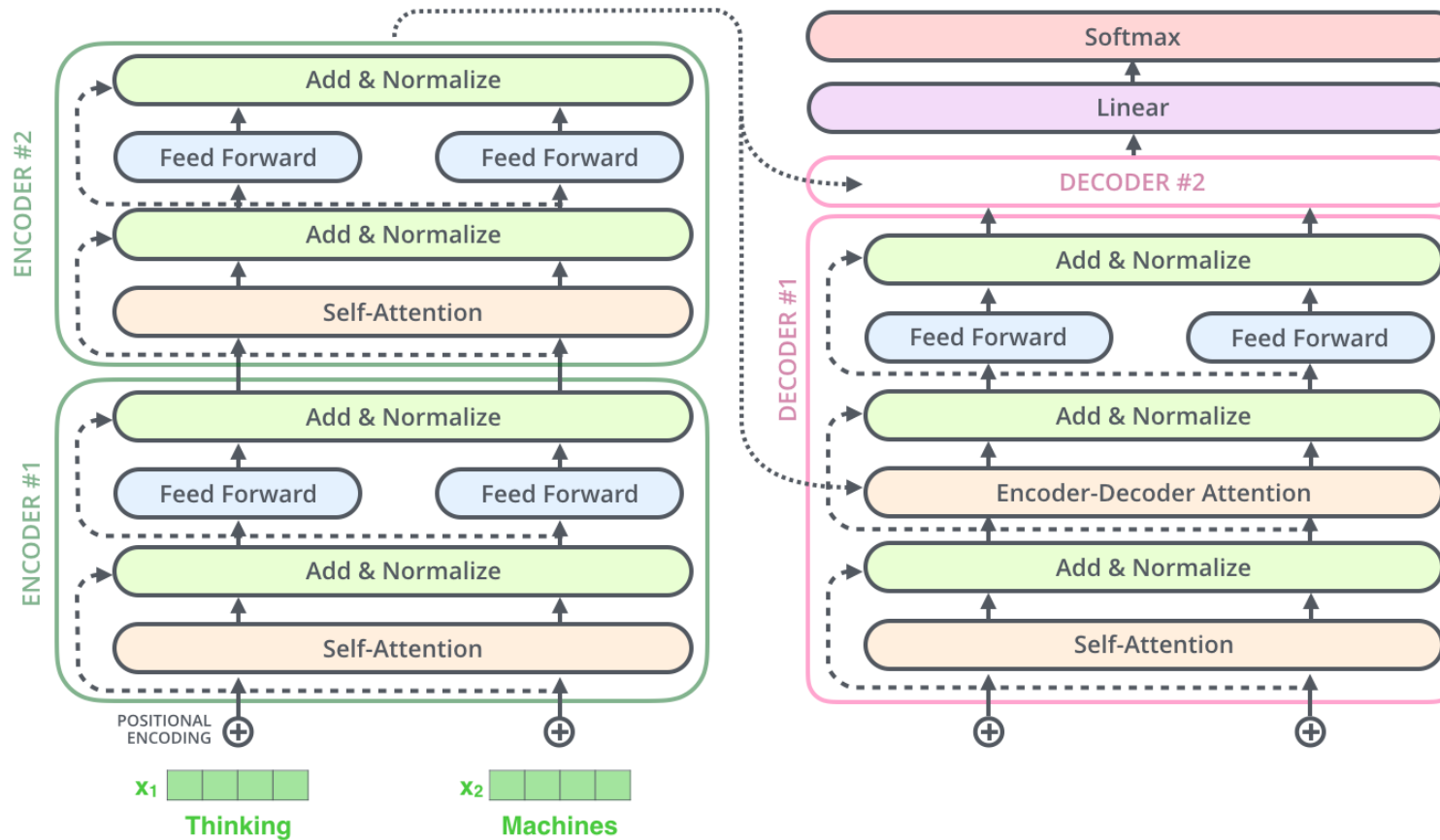
computes a representation of the sequence

can more easily **capture long-distance dependencies** compared to a RNN

calculation can be **parallelized**

[1] Vaswani et al. : Attention is all you need, NIPS 2017

Deep Neural Network – Transformer Architecture

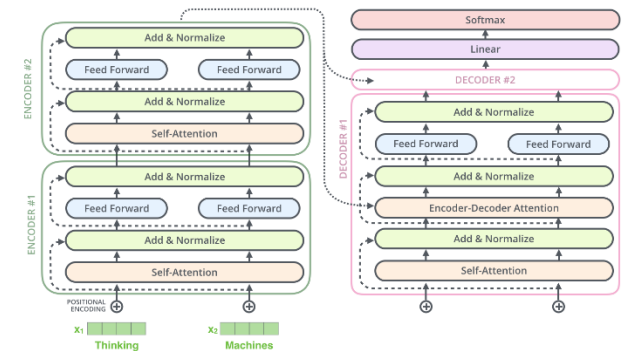
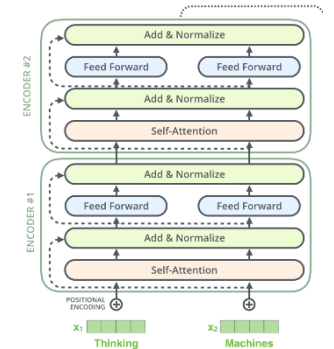
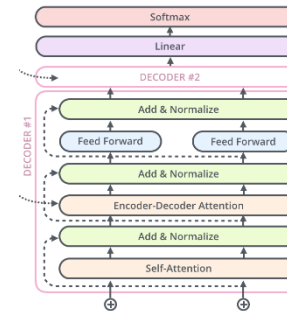


<https://jalammar.github.io/illustrated-transformer/>

Tutorial: <http://nlp.seas.harvard.edu/annotated-transformer/>

Applications and Models

- **Encoder Models**
e.g., **Bert** model [1], used for e.g. text **classification**, **named entity recognition**
- **Decoder Models**
e.g., GPT-X [2], used for e.g. text **generation**
- **Encoder-Decoder Models**
e.g., T5 [3], used for e.g. translation tasks



Explanation and implementation: <https://huggingface.co/learn/nlp-course/>

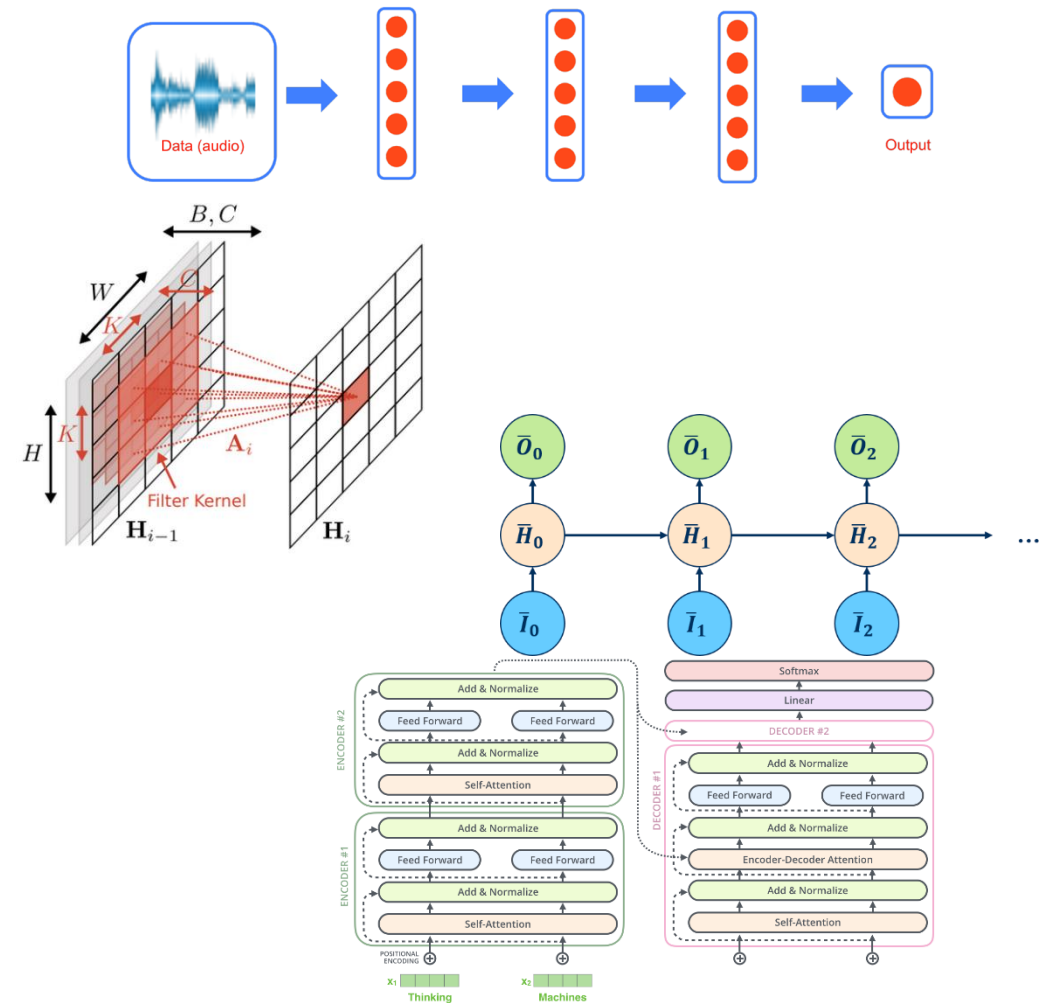
[1] Devlin et al.: BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding, NAACL,HLT 2019

[2] Radford and Sutskever: Improving Language Understanding by Generative Pre-Training, arxiv, 2018

[3] Raffel et al.: Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer, Journal of Machine Learning Research, 2020

Deep Learning Architectures and Applications and - Takeaway

- Deep Neural Networks allow to learn processing steps, which has been hand-crafted before
- Convolutional Neural Networks allow to share weight and thus reduce number of parameters
- Applied for image processing
- Recurrent Neural Networks allow to process sequences and take history into account
- Transformer based architectures are SotA for NLP
 - „Attention is all you need“
 - Encoder, Decoder and Encoder-Decoder models depending on NLP task



Summary - Machine Learning

- Introduction to Machine Learning
- Similarity and Distances
- Clustering
- Supervised Learning – Training and Learning
- Classification
 - Decision tree
 - Bayes Classification Methods
- Model Evaluation and Selection
- Dimension Reduction
- Linear Regression
- Logistic Regression
- Deep learning



Further References

Convolutional Neural Networks and Recurrent Neural Networks

- <https://uni-tuebingen.de/fakultaeten/mathematisch-naturwissenschaftliche-fakultaet/fachbereiche/informatik/lehrstuehle/autonomous-vision/lectures/deep-learning/>

Lecture 7 Convolutional Neural Networks and **Lecture 08** Sequence Models

Goodfellow, I. et al., *Deep Learning*, MIT Press, 2015

- <https://github.com/janishar/mit-deep-learning-book-pdf>

Deep Learning Lecture Prof. Andreas Geiger, University Tübingen

- <https://uni-tuebingen.de/fakultaeten/mathematisch-naturwissenschaftliche-fakultaet/fachbereiche/informatik/lehrstuehle/autonomous-vision/lectures/deep-learning/>

Dive into Deep learning

- <http://d2l.ai/index.html>

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